

SOFTWARE ENGINEERING TECHNICAL DOCUMENTATION

SOFTWARE REQUIREMENTS SPECIFICATION

Automated Library Management System (ALMS)

Document Standard: IEEE Std 830-1998	Architecture: 3-Tier Client-Server	Persistence: Relational DBMS (SQL)	Security Level: TLS 1.3 / Role-Based Access
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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) establishes the formal functional, behavioral, performance, and infrastructural specifications for the Automated Library Management System (ALMS). It serves as the primary technical specification for developers, database architects, system testers, project supervisors, and administrative personnel during the development, verification, and deployment lifecycle.

1.2 Scope

The Automated Library Management System is a centralized information and inventory management solution engineered to automate circulation workflows, bibliographic record tracking, and patron interactions within public and institutional libraries.

System Boundary Overview:

- In-Scope Modules:** Bibliographic catalog management (books, journals, reference media), barcode-driven physical copy inventory, patron membership lifecycle management, circulation counters (issue, return, renewal), fine calculation and receipt ledgering, reservation queueing (holds), and Online Public Access Catalog (OPAC) multi-parameter search.
- Out-of-Scope Modules:** Full-text automated document scanning and optical character recognition (OCR), electronic book DRM reader engines, and logistics tracking for external postal courier consignments.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
ALMS	Automated Library Management System
API	Application Programming Interface
Barcode / QR	Optical machine-readable data format applied to books and borrower ID cards
CRUD	Create, Read, Update, and Delete operations
DBMS	Database Management System
ISBN	International Standard Book Number (10 or 13 numeric digits)
JWT	JSON Web Token for stateless, signed session authentication
OPAC	Online Public Access Catalog
Patron	Registered borrower holding an active borrowing membership card
RBAC	Role-Based Access Control
SRS	Software Requirements Specification

1.4 References

1. IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society.

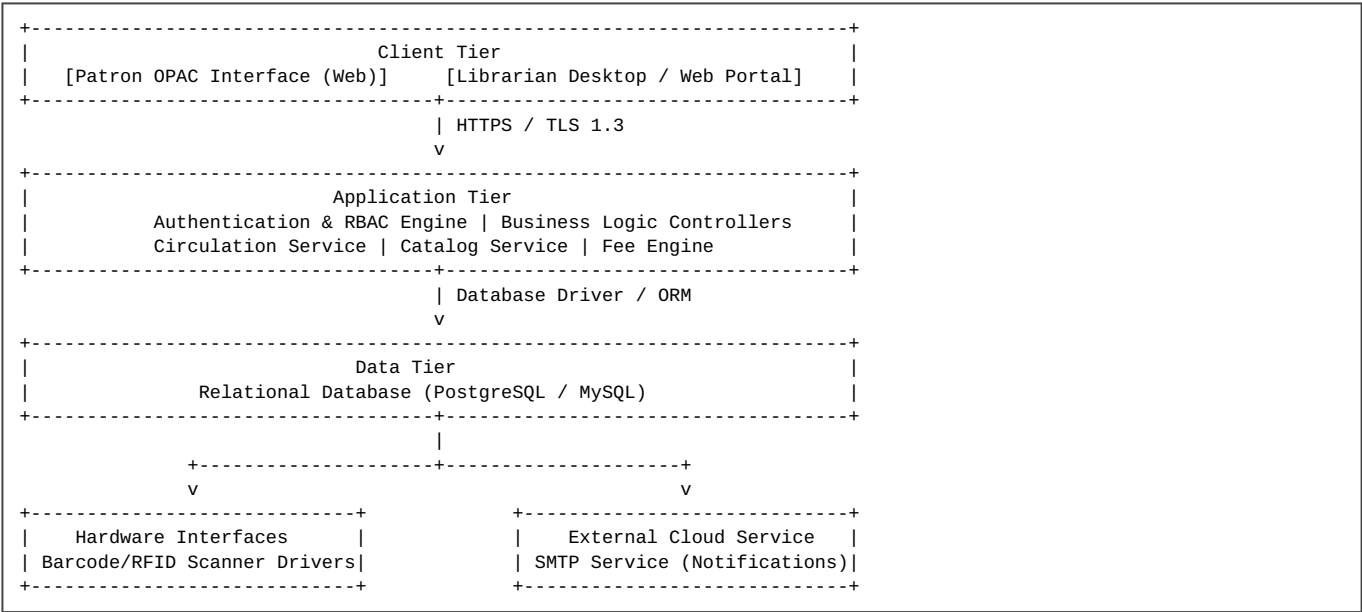
2. ISO/IEC/IEEE 29148:2018: *Systems and software engineering — Life cycle processes — Requirements engineering*.

3. ISO/IEC 27001:2022: *Information security, cybersecurity and privacy protection — Information security management systems*.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system operates as an independent, multi-tier database-backed application. The presentation layer consists of web interfaces for public discovery and administrative desktop interfaces for circulation desks. The application tier executes business logic and communicates securely with relational database servers, local USB barcode scanners, thermal receipt printers, and SMTP alert dispatchers.



2.2 User Classes and Characteristics

- **Guest User:** General public visiting the OPAC portal to search titles and inspect live shelf availability without reservation or loan rights.
- **Patron:** Registered member (student, academic, or general borrower). Authorized to access personal loan logs, request loan renewals, place hold requests on borrowed books, and track fine balances.
- **Circulation Librarian:** Staff user at the circulation desk. Operates barcode scanner hardware to process checkouts, process check-ins, record fine payments, and manage physical inventory condition.
- **Catalog Administrator:** Chief librarian or technical administrator responsible for catalog entry creation, bulk record imports (MARC/CSV), policy parameter updates (fine rates, borrow limits), and staff authorization.

2.3 Operating Environment

- **Server Hardware:** Quad-core x86-64 processor, 8 GB RAM, 100 GB SSD storage.
- **Server OS:** Linux (Ubuntu Server 22.04 LTS or RHEL 9) or Windows Server 2022.
- **Client Browsers:** Chromium 105+, Firefox 100+, Safari 16+ supporting responsive display rendering at \$1280 imes 720\$ and higher.
- **Database:** PostgreSQL 14+ or MySQL 8.0+.

3. FUNCTIONAL REQUIREMENTS

3.1 User Authentication and Authorization

- **FR-01:** The system shall authenticate Patrons and Staff using unique identifiers paired with salted, hashed passwords (Argon2id or BCrypt).
- **FR-02:** The system shall enforce Role-Based Access Control (RBAC) ensuring only authorized staff accounts can execute circulation and catalog-mutation operations.
- **FR-03:** The system shall automatically terminate authenticated user sessions after 15 minutes of inactivity.

3.2 Catalog and Inventory Management

- **FR-04:** The system shall allow administrators to create, read, update, and archive master bibliographic entries (Title, Author, ISBN, Publisher, Year, Edition, Subject Category, and Shelf Location).
- **FR-05:** The system shall track physical copies by linking multiple unique copy barcode IDs to a single master bibliographic record.
- **FR-06:** The system shall prohibit the deletion of any catalog entry or physical copy that currently has an active loan status.

3.3 Catalog Search (OPAC)

- **FR-07:** The system shall provide an OPAC search function supporting multi-field queries across Title, Author, ISBN, Category, and Keywords.
- **FR-08:** The system shall return real-time item status (Available, Checked Out, Reserved, Maintenance) along with shelf call numbers.

3.4 Circulation Operations

- **FR-09:** The system shall allow librarians to execute book checkout upon scanning the patron ID card and the copy barcode, enforcing configured loan limits (e.g., 5 books for students, 10 for faculty).
- **FR-10:** The system shall block checkout transactions if the patron has reached maximum allowed active loans, possesses overdue items, or holds unpaid fines exceeding the configured balance threshold.
- **FR-11:** The system shall register book returns upon scanning the copy barcode, mark the copy as Available, and automatically compute fine balances if the transaction timestamp exceeds the due date.
- **FR-12:** The system shall allow a patron to renew an active loan once, provided no other patron has placed an active hold on that title.

3.5 Hold and Reservation Engine

- **FR-13:** The system shall allow authenticated patrons to place a reservation hold on a title if all physical copies are currently checked out.
- **FR-14:** The system shall maintain hold reservations using a strict First-In, First-Out (FIFO) queue.
- **FR-15:** When a reserved book is checked in, the system shall mark the item as Reserved, retain it at the circulation desk for 48 hours, and dispatch an automated email notice to the next patron in the queue.

3.6 Fines and Payments

- **FR-16:** The system shall calculate daily overdue fines using the configured rule: $\text{Fine} = (\text{Days Overdue}) \times (\text{Daily Rate})$.
- **FR-17:** The system shall record fine payments, print/email transaction receipts, and deduct the paid sum from the patron's account balance.

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

- **NFR-01 (Search Latency):** OPAC search queries shall execute and display results in $\leq 1.0 \text{ second}$ under nominal loads of 500 concurrent users.
- **NFR-02 (Circulation Throughput):** Barcode scan and checkout processing operations shall complete within $\leq 500 \text{ milliseconds}$.
- **NFR-03 (Transaction Concurrency):** The database engine shall support a minimum sustained throughput of 100 transactions per second without deadlocks.

4.2 Security Requirements

- **NFR-04 (Transport Security):** All network communications between clients and application servers must enforce TLS 1.3 encryption over HTTPS.
- **NFR-05 (Credential Protection):** Passwords must be hashed using Argon2id or BCrypt with individual salts. Plaintext passwords must never be stored or written to log files.
- **NFR-06 (Injection Prevention):** All database access must use parameterized queries or Prepared Statements via an Object-Relational Mapper (ORM) to eliminate SQL Injection (SQLi) vulnerabilities.

4.3 Reliability and Availability

- **NFR-07 (System Uptime):** The system shall maintain an operational availability of 99.5%, excluding scheduled maintenance windows.
- **NFR-08 (Data Protection):** Automated relational database backups shall occur daily with an operational Recovery Point Objective (RPO) $\leq 24 \text{ hours}$ and Recovery Time Objective (RTO) $\leq 2 \text{ hours}$.

4.4 Usability and Accessibility

- **NFR-09 (Workflow Efficiency):** Front-desk issue and return workflows shall be executable by a trained operator in 4 or fewer input steps.
- **NFR-10 (Accessibility Standards):** The public OPAC interface shall comply with WCAG 2.1 Level AA accessibility standards.

5. DESIGN CONSTRAINTS

- **DC-01 (Architectural Separation):** The system must adhere to a strict 3-tier architecture maintaining modular isolation between the presentation layer, business logic layer, and persistence store.
- **DC-02 (Bibliographic Standards):** Catalog records must accommodate both 10-digit and 13-digit ISBN structures and Dublin Core bibliographic metadata definitions.
- **DC-03 (Hardware Protocol):** Barcode scanners must operate using standard USB Human Interface Device (HID) keyboard-wedge emulation to prevent proprietary device driver dependency.

6. ASSUMPTIONS AND DEPENDENCIES

- **Assumption 1:** Circulation desk terminals maintain continuous local area network (LAN) access to the central server.
- **Assumption 2:** Library media are labeled with Code 128 or QR barcode tags prior to being shelved for public access.
- **Dependency 1:** Automated email notifications depend upon external SMTP server uptime and network deliverability.
- **Dependency 2:** Workstation clients run modern web browsers with standard JavaScript execution enabled.

7. DETAILED SYSTEM USE CASES

Use Case UC-01: Book Checkout

Primary Actor: Circulation Librarian | **Secondary Actor:** Patron

Preconditions: Librarian is authenticated; Patron membership is active without hold blocks or excessive unpaid fines; physical copy status is Available.

Main Success Scenario:

1. Librarian scans Patron barcode; system displays borrower profile, loan count, and fine balance.
2. Librarian scans the barcode affixed to the physical book.
3. System validates that the book is available and that the patron has not exceeded the borrowing limit.
4. System records a new loan transaction with the current timestamp and computes the scheduled due date.
5. System updates the copy status to Checked Out and issues a loan confirmation slip.

Alternate Flows:

- *Patron Ineligible:* If overdue books exist or unpaid fines exceed policy limits, checkout is halted and an error notification is displayed.
- *Item Unavailable:* If the item is marked as Checked Out or Reserved for another borrower, the system flags the conflict and terminates the checkout.

Postconditions: Loan record logged, copy status set to Checked Out, patron active loan counter incremented.

Use Case UC-02: Book Return

Primary Actor: Circulation Librarian

Preconditions: Physical book is presented at the counter and is recorded as Checked Out.

Main Success Scenario:

1. Librarian scans the book copy barcode; system retrieves the corresponding active loan record.
2. System checks return timestamp against the due date; verified as returned on or before the due date.
3. System closes the loan record with the return timestamp and marks copy status as Available.
4. System checks the reservation queue; finding no holds, the item is marked for re-shelving.

Alternate Flows:

- *Overdue Return:* If returned after the due date, system calculates overdue fine, appends it to the patron ledger, and alerts the operator.
- *Reserved Item:* If an active reservation exists, system marks the item as Reserved, holds it for 48 hours, and triggers an automated notification email to the next patron in the queue.

Postconditions: Active loan record closed, copy status updated to Available or Reserved, fine recorded if overdue.

Use Case UC-03: Reserve a Title (Place Hold)

Primary Actor: Patron

Preconditions: Patron is authenticated; all physical copies of the requested title are currently Checked Out.

Main Success Scenario:

1. Patron searches for title on OPAC and views detail screen.
2. Patron clicks "Place Hold / Reserve".
3. System validates patron eligibility (active account status, hold limit compliance).
4. System appends the patron to the end of the FIFO hold queue for that bibliographic record.
5. System displays reservation confirmation and current queue position.

Alternate Flows:

- *Available Copy Exists:* System rejects hold request and displays shelf location for in-person collection.
- *Duplicate Hold:* If patron already holds an active reservation on this title, request is rejected.

Postconditions: Reservation entry registered in FIFO queue linked to patron and title.

Use Case UC-04: Search Catalog (OPAC)

Primary Actor: Guest User / Patron

Preconditions: Public OPAC interface access.

Main Success Scenario:

1. User inputs search terms into the OPAC query field and selects optional filters (Author, Category, Year).
2. User submits search request; system executes parameterized search across catalog indices.
3. System displays matching bibliographic records with Title, Author, Year, Subject, and live Availability summary.
4. User selects a record to inspect full details and physical shelf location.

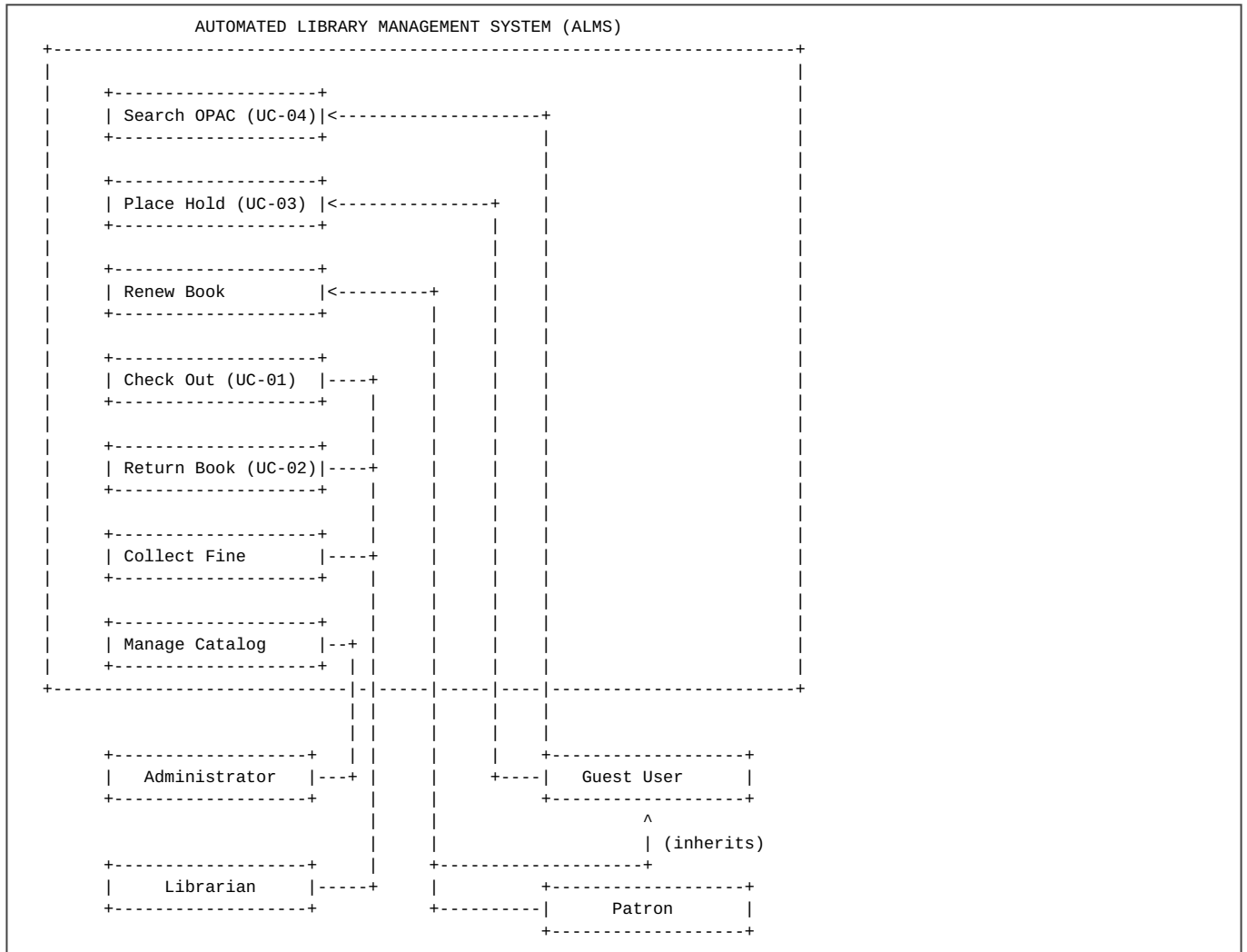
Alternate Flows:

- *No Matches:* If query yields no matches, system displays a notification suggesting refined keywords.

Postconditions: Bibliographic query results displayed without altering database state.

8. SUPPORTING UML DIAGRAMS

8.1 Use Case Diagram



8.2 Domain Class Diagram

The structural class model below depicts the core operational entities, associations, and cardinalities governing the library system:

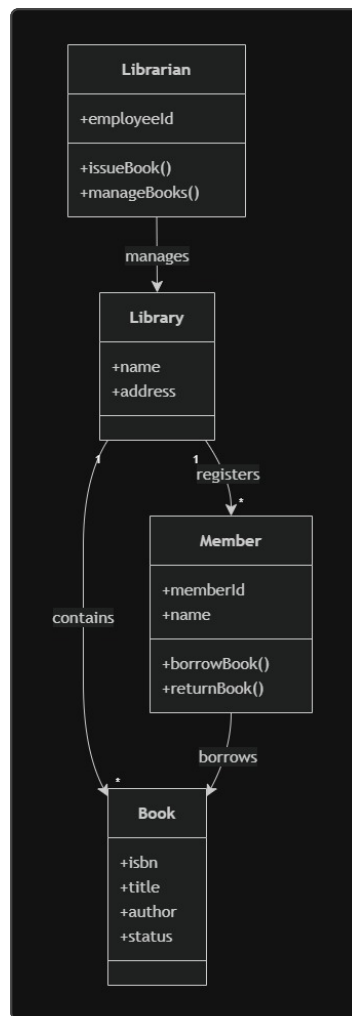
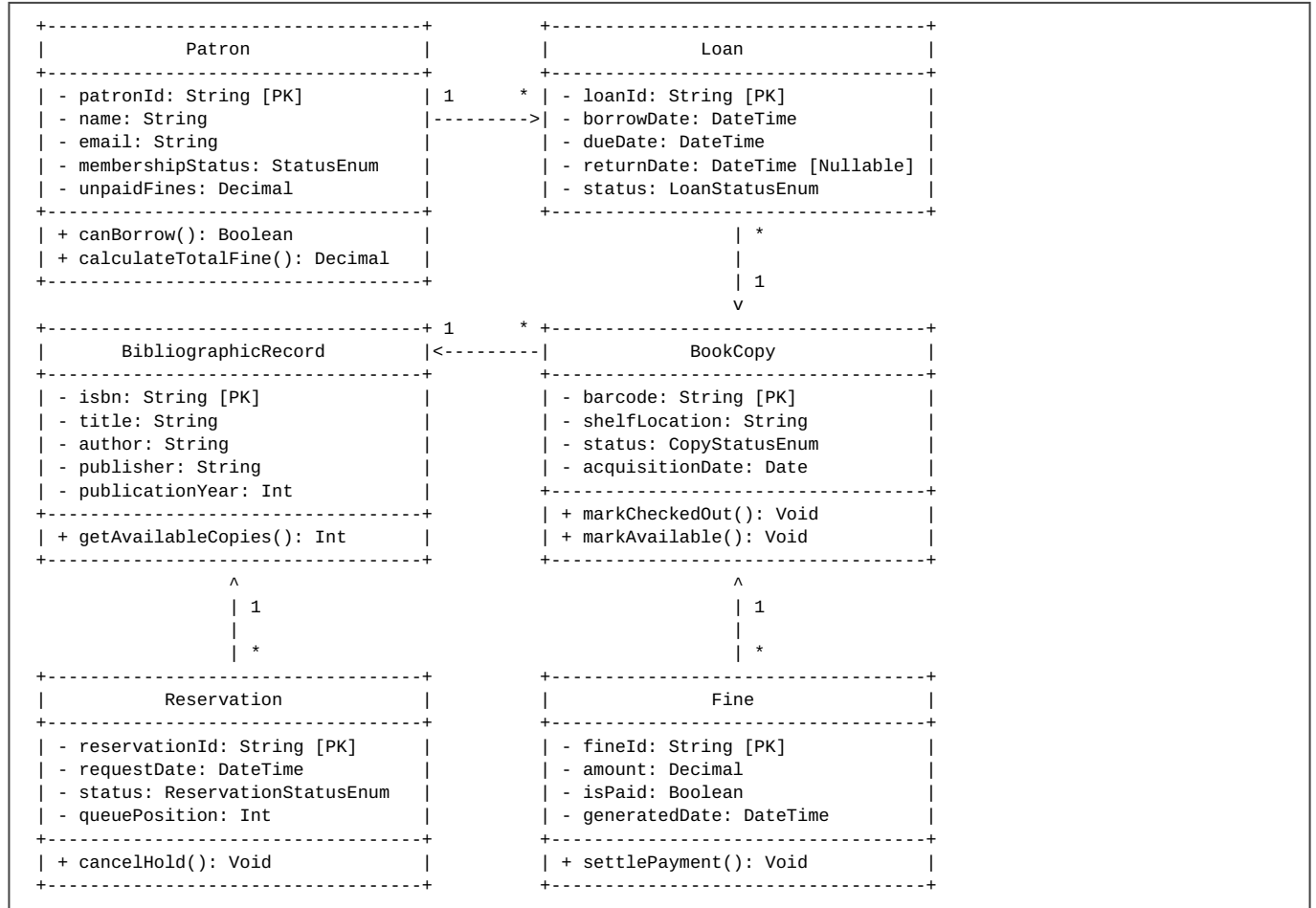


Figure 8.1: Core Library Management System Class Diagram

8.3 Extended Relational Domain Model



8.4 Sequence Diagram: Book Checkout Workflow (UC-01)

